https://github.com/popescoup/Hw5_DATA226

1)

```
(base) lucapopescu@Lucas-MacBook-Pro ~ % spark-submit --version
26/09/28 22:32:13 WARN Utils: Your hostname, Lucas-MacBook-Pro.local resolves to a loop
back address: 127.0.0.1; using 192.168.1.243 instead (on interface en0)
26/09/28 22:32:13 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.5.8
      /_/

Using Scala version 2.12.18, OpenJDK 64-Bit Server VM, 17.0.20.1
Branch HEAD
Compiled by user runner on 2026-01-12T04:16:44Z
Revision 5a48a37b2dbd7b51e3640cd1d947438459556cc6
Url https://github.com/apache/spark
Type --help for more information.
(base) lucapopescu@Lucas-MacBook-Pro ~ % date
Mon Sep 28 22:32:24 PDT 2026
```

2)

```
(base) lucapopescu@Lucas-MacBook-Pro sjsu-data226-FA26 % git pull
Already up to date.
(base) lucapopescu@Lucas-MacBook-Pro sjsu-data226-FA26 % cd week6
(base) lucapopescu@Lucas-MacBook-Pro week6 % ls -lh
total 100424
-rw-r--r--  1 lucapopescu  staff   1.5K Sep 28 22:47 log_file_handling_in_spark.py
-rw-r--r--  1 lucapopescu  staff   1.3K Sep 28 22:47 pyspark_port.py
-rw-r--r--  1 lucapopescu  staff     2B Sep 28 22:47 sample_web_log_1.log.gz
-rw-r--r--  1 lucapopescu  staff     2B Sep 28 22:47 sample_web_log_2.log.gz
-rw-r--r--  1 lucapopescu  staff   9.8M Sep 28 22:47 sample_web_log_3.log.gz
-rw-r--r--  1 lucapopescu  staff   9.8M Sep 28 22:47 sample_web_log_4.log.gz
-rw-r--r--  1 lucapopescu  staff   9.8M Sep 28 22:47 sample_web_log_5.log.gz
-rw-r--r--  1 lucapopescu  staff   9.8M Sep 28 22:47 sample_web_log_6.log.gz
-rw-r--r--  1 lucapopescu  staff   9.8M Sep 28 22:47 sample_web_log_7.log.gz
-rw-r--r--  1 lucapopescu  staff   257B Sep 28 22:47 test_spark.py
```

3)

```
(base) lucapopescu@Lucas-MacBook-Pro week6 % spark-submit hw5.py
26/09/28 23:01:06 WARN Utils: Your hostname, Lucas-MacBook-Pro.local resolves to a loopback
 address: 127.0.0.1; using 192.168.1.243 instead (on interface en0)
26/09/28 23:01:06 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
26/09/28 23:01:07 INFO SparkContext: Running Spark version 3.5.8
```

6

```
+--------------------------------------------------------------------------------------+
|value                                                                                 |
+--------------------------------------------------------------------------------------+
|123.45.67.89 - - [05/Nov/2024:02:08:16 +0000] "DELETE /cart HTTP/1.1" 500 242        |
|192.168.1.1 - - [04/Nov/2024:21:23:39 +0000] "POST /checkout HTTP/1.1" 404 2781      |
|234.56.78.90 - - [05/Nov/2024:07:06:19 +0000] "GET /api/data HTTP/1.1" 301 3758      |
|192.168.1.1 - - [04/Nov/2024:20:03:56 +0000] "POST /home HTTP/1.1" 200 1837          |
|192.168.1.1 - - [04/Nov/2024:21:25:05 +0000] "GET /products/123 HTTP/1.1" 200 3430   |
|234.56.78.90 - - [04/Nov/2024:07:38:10 +0000] "GET /api/data HTTP/1.1" 404 3729      |
|123.45.67.89 - - [04/Nov/2024:12:33:22 +0000] "PUT /api/data HTTP/1.1" 404 799       |
|192.168.1.1 - - [04/Nov/2024:07:37:46 +0000] "GET /api/data HTTP/1.1" 500 309        |
|123.45.67.89 - - [04/Nov/2024:21:52:36 +0000] "POST /checkout HTTP/1.1" 301 2375     |
|123.45.67.89 - - [04/Nov/2024:08:36:44 +0000] "DELETE /api/data HTTP/1.1" 404 3449   |
|192.168.1.1 - - [05/Nov/2024:03:15:43 +0000] "GET /api/data HTTP/1.1" 200 2319       |
|234.56.78.90 - - [05/Nov/2024:01:26:03 +0000] "DELETE /home HTTP/1.1" 500 1168       |
|234.56.78.90 - - [05/Nov/2024:03:26:33 +0000] "DELETE /cart HTTP/1.1" 500 1262       |
|123.45.67.89 - - [04/Nov/2024:20:46:25 +0000] "PUT /home HTTP/1.1" 301 4401          |
|123.45.67.89 - - [05/Nov/2024:08:07:51 +0000] "GET /api/data HTTP/1.1" 301 3736      |
|123.45.67.89 - - [04/Nov/2024:21:01:30 +0000] "DELETE /cart HTTP/1.1" 404 2418       |
|123.45.67.89 - - [04/Nov/2024:09:40:29 +0000] "POST /api/data HTTP/1.1" 301 3260     |
|234.56.78.90 - - [04/Nov/2024:09:23:42 +0000] "GET /home HTTP/1.1" 200 1488          |
|192.168.1.1 - - [04/Nov/2024:11:53:57 +0000] "POST /products/123 HTTP/1.1" 200 2627  |
|234.56.78.90 - - [05/Nov/2024:01:26:01 +0000] "PUT /cart HTTP/1.1" 500 4406          |
+--------------------------------------------------------------------------------------+
only showing top 20 rows
```

```
+-----------+--------------------+------+-------------+------+----+
|         ip|           timestamp|method|          url|status|size|
+-----------+--------------------+------+-------------+------+----+
|123.45.67.89|05/Nov/2024:02:08...|DELETE|        /cart|   500| 242|
| 192.168.1.1|04/Nov/2024:21:23...|  POST|    /checkout|   404|2781|
|234.56.78.90|05/Nov/2024:07:06...|   GET|    /api/data|   301|3758|
| 192.168.1.1|04/Nov/2024:20:03...|  POST|        /home|   200|1837|
| 192.168.1.1|04/Nov/2024:21:25...|   GET|/products/123|   200|3430|
|234.56.78.90|04/Nov/2024:07:38...|   GET|    /api/data|   404|3729|
|123.45.67.89|04/Nov/2024:12:33...|   PUT|    /api/data|   404| 799|
| 192.168.1.1|04/Nov/2024:07:37...|   GET|    /api/data|   500| 309|
|123.45.67.89|04/Nov/2024:21:52...|  POST|    /checkout|   301|2375|
|123.45.67.89|04/Nov/2024:08:36...|DELETE|    /api/data|   404|3449|
| 192.168.1.1|05/Nov/2024:03:15...|   GET|    /api/data|   200|2319|
|234.56.78.90|05/Nov/2024:01:26...|DELETE|        /home|   500|1168|
|234.56.78.90|05/Nov/2024:03:26...|DELETE|        /cart|   500|1262|
|123.45.67.89|04/Nov/2024:20:46...|   PUT|        /home|   301|4401|
|123.45.67.89|05/Nov/2024:08:07...|   GET|    /api/data|   301|3736|
|123.45.67.89|04/Nov/2024:21:01...|DELETE|        /cart|   404|2418|
|123.45.67.89|04/Nov/2024:09:40...|  POST|    /api/data|   301|3260|
|234.56.78.90|04/Nov/2024:09:23...|   GET|        /home|   200|1488|
| 192.168.1.1|04/Nov/2024:11:53...|  POST|/products/123|   200|2627|
|234.56.78.90|05/Nov/2024:01:26...|   PUT|        /cart|   500|4406|
+-----------+--------------------+------+-------------+------+----+
only showing top 20 rows
```

4)

```
# Step 4: (status, method) counts using DataFrame operations
status_method_count = (
    log_df
    .filter(log_df.method != "DELETE")
    .groupBy("status", "method")
    .count()
    .orderBy(F.asc("count"))
)
status_method_count.show()
```

```
+------+------+------+
|status|method| count|
+------+------+------+
|   404|   PUT|311787|
|   301|   PUT|311820|
|   301|   GET|312133|
|   200|   PUT|312581|
|   500|   PUT|312606|
|   404|  POST|312606|
|   500|  POST|312703|
|   200|  POST|312897|
|   200|   GET|313000|
|   404|   GET|313058|
|   301|  POST|313140|
|   500|   GET|313945|
+------+------+------+
```

5)

```
# Step 5: (status, method) counts using Spark SQL
log_df.createOrReplaceTempView("logs")

status_method_count_sql = spark.sql("""
SELECT status, method, COUNT(1) AS count
FROM logs
WHERE method != 'DELETE'
GROUP BY status, method
ORDER BY count ASC
""")
status_method_count_sql.show()
```

```
+------+------+------+
|status|method| count|
+------+------+------+
|   404|   PUT|311787|
|   301|   PUT|311820|
|   301|   GET|312133|
|   200|   PUT|312581|
|   500|   PUT|312606|
|   404|  POST|312606|
|   500|  POST|312703|
|   200|  POST|312897|
|   200|   GET|313000|
|   404|   GET|313058|
|   301|  POST|313140|
|   500|   GET|313945|
+------+------+------+
```